

Note on: *CDS Market Formulas and Models*

Damiano Brigo & Massimo Morini, 2005

Reading notes

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One-paragraph summary

The paper builds a Swap Market Model analogue for credit derivatives: a CDS Market Model in which the par CDS spread $R_{a,b}(t)$ is the natural underlying, with a Black-Scholes-type closed-form pricing formula for European CDS options under a lognormal-spread assumption. The construction uses the Jeanblanc-Rutkowski subfiltration setup — $\mathcal{F}_t = \mathcal{F}_t^r \vee \mathcal{H}_t$, separating default information from default-free information — to keep all measures equivalent to the risk-neutral measure (avoiding the defaultable-bond numeraire pathology in Schonbucher 1999). The defaultable PV01 $C_{a,b}(t) = \sum_i \alpha_i \bar{P}(t, T_i)$ serves as numeraire, defining a measure $\bar{Q}^{a,b}$ under which the par spread is a martingale, yielding the Black formula (4). Under the conditional-independence assumption between the subfiltration $\mathcal{H}_t$ and the risk-neutral measure, the framework also allows extension to a CDS term-structure model by parameterising one-period rates $R_j(t)$, with the practical complication that defaultable-bond numeraire ratios cannot be expressed purely in terms of R_j unless one further assumes independence of default-free interest rates and default probabilities. Under this independence, a full LMM-style CDS market model is defined and applied to constant-maturity CDS (CMCDS) via the drift-freezing approximation; a closed-form CMCDS price is obtained. Empirical examples on European corporate CDS options (Deutsche Telecom, Daimler Chrysler, France Telecom) and a CMCDS on Fiat are included.

1 Setup and notation

Paper	House	Meaning
τ	τ	Default time of reference entity
$T_a, \dots, T_b$	T_i	Payment dates of the CDS premium leg
α_i	α_i	Year fraction $T_{i-1} \rightarrow T_i$
LGD	$1 - \text{Rec}$	Loss given default
R	R	Running CDS premium / spread paid by buyer
$R_{a,b}(t)$	$S^{\text{CDS}}(t)$	Par (fair) CDS spread, maturity $T_a \rightarrow T_b$
$D(t, T)$	$D_{t,T}$	Stochastic bank-account discount B_t/B_T
$P(t, T)$	$D_{t,T}$	Default-free zero-coupon bond
$\bar{P}(t, T)$	—	Defaultable zero (pre-default) = $\mathbb{E}^Q[D(t, T) \mathbf{1}_{\tau > T} \mathcal{H}_t] / Q(\tau > t \mathcal{H}_t)$
$Q(\tau > t \mathcal{H}_t)$	$Q(t, T)$	Conditional survival probability
$\mathcal{F}_t$	—	Full filtration (default time visible)
$\mathcal{H}_t$	—	Default-free subfiltration
$\mathcal{F}_t^\tau$	—	Subfiltration generated by τ
Q	$\mathbb{Q}$	Risk-neutral measure
$\bar{Q}^{a,b}$	—	Measure with numeraire $C_{a,b}$ (defaultable PV01)
$C_{a,b}(t)$	$\text{PV01}_{a,b}^{\text{Risky}}(t)$	Defaultable PV01 $\sum_i \alpha_i Q(\tau > t \mathcal{H}_t) \bar{P}(t, T_i)$
Q^j	—	T_j -forward measure of default-free bond $P(t, T_j)$
$Q^j(\tau > T \mathcal{H}_t)$	—	Forward conditional survival
$F_k(t)$	—	Default-free forward Libor (LMM variable)
$R_j(t) = R_{j-1,j}(t)$	—	One-period forward CDS spread
$\bar{\sigma}_{a,b}$	—	Implied volatility of $R_{a,b}$ under $\bar{Q}^{a,b}$
Black(F, K, v)	—	Standard Black formula

Subfiltration setup. The full σ -algebra is $\mathcal{F}_t = \mathcal{F}_t^\tau \vee \mathcal{H}_t$: full information equals knowledge of the default time plus knowledge of all default-free observables (rates, market quotes). The point is that the conditional survival probability $Q(\tau > t | \mathcal{H}_t)$ is strictly positive in every state of the world, while $Q(\tau > t | \mathcal{F}_t) = \mathbf{1}_{\tau > t}$ jumps to zero at default. Working under $\mathcal{H}_t$ keeps the numeraire $C_{a,b}$ strictly positive, hence all change-of-measure machinery works.

2 Key equations

CDS payoff convention (paper's eq. 1, slightly more tractable than the textbook payment-at-default):

$$\text{CDS}\Pi_t(R) = \text{LGD} \sum_{i=a+1}^b D(t, T_i) \mathbf{1}_{T_{i-1} < \tau \leq T_i} - \sum_{i=a+1}^b D(t, T_i) \alpha_i \mathbf{1}_{\tau > T_i} R. \quad (1)$$

Says: the protection leg pays LGD at the first T_i following default (not at τ itself — the “postponed-default” simplification); the premium leg pays R times α_i at each T_i if alive, with *no* accrued premium on the default-period stub.

Par CDS spread (paper's central definition, with subfiltration):

$$R_{a,b}(t) = \text{LGD} \cdot \frac{\sum_{i=a+1}^b \mathbb{E}^Q[D(t, T_i) \mathbf{1}_{T_{i-1} < \tau \leq T_i} | \mathcal{H}_t]}{\sum_{i=a+1}^b \alpha_i Q(\tau > t | \mathcal{H}_t) \bar{P}(t, T_i)}, \quad (2)$$

with $\bar{P}(t, T) = \mathbb{E}^Q[D(t, T) \mathbf{1}_{\tau > T} | \mathcal{H}_t] / Q(\tau > t | \mathcal{H}_t)$ the pre-default defaultable bond price. *Says:* the par spread is a ratio of expected protection payments to the defaultable annuity. Using $\mathcal{H}_t$ conditioning rather than $\mathcal{F}_t$ avoids zero denominators at default.

Defaultable PV01 numeraire:

$$C_{a,b}(t) := \sum_{i=a+1}^b \alpha_i Q(\tau > t | \mathcal{H}_t) \bar{P}(t, T_i) = \sum_{i=a+1}^b \alpha_i \mathbb{E}^Q[D(t, T_i) \mathbf{1}_{\tau > T_i} | \mathcal{H}_t]. \quad (3)$$

Says: the defaultable PV01 (“DV01 of one bp of premium”) is strictly positive (Proposition 7 of the paper); hence its accreted version $C_{a,b} = C_{a,b}(T_a) B/B_{T_a}$ is a valid numeraire, defining the measure $\bar{Q}^{a,b}$ equivalent to Q .

Standard Market Formula for CDS options (paper’s eq. 8):

$$\text{CDSOption}_t = \mathbf{1}_{\tau > t} \sum_{i=a+1}^b \alpha_i \bar{P}(t, T_i) \text{Black}(R_{a,b}(t), K, \bar{\sigma}_{a,b} \sqrt{T_a - t}), \quad (4)$$

with $\text{Black}(F, K, v) = F N(d_1) - K N(d_2)$, $d_{1,2} = [\ln(F/K) \pm \frac{1}{2}v^2]/v$, under the lognormal assumption $dR_{a,b}(t) = \bar{\sigma}_{a,b} R_{a,b}(t) dV^{a,b}$ where $V^{a,b}$ is Brownian motion under $\bar{Q}^{a,b}$. *Says:* the CDS option prices like a Black caplet/swaption, with the par CDS spread as the underlying and the defaultable PV01 as the discounted notional. The novelty is that, thanks to the subfiltration setup, $\bar{Q}^{a,b}$ is equivalent to Q (not absolutely continuous only), so implied vols are well-defined.

CDS spread as forward survival-probability ratio (paper’s eq. 13, central one-period structural identity, under $\mathcal{H}$ -conditional independence):

$$R_j(t) = \frac{\text{LGD}}{\alpha_j} \left(\frac{Q^j(\tau > T_{j-1} | \mathcal{H}_t)}{Q^j(\tau > T_j | \mathcal{H}_t)} - 1 \right). \quad (5)$$

Says: a one-period CDS spread is a ratio of forward conditional survival probabilities under the T_j -forward measure, with default-free discount factors completely cancelled — analogous to $F_k(t) = (P(t, T_{k-1})/P(t, T_k) - 1)/\alpha_k$ for a Libor forward. This is the key conceptual result: *the CDS spread is fundamentally a probability ratio, not a rate.*

Multi-period spread as weighted one-period combination (paper’s eq. 10):

$$R_{a,b}(t) = \sum_{j=a+1}^b \frac{\alpha_j \bar{P}(t, T_j)}{\sum_i \alpha_i \bar{P}(t, T_i)} R_j(t). \quad (6)$$

Says: the term-structure CDS spread is a defaultable-PV01-weighted average of one-period spreads, exact analogue of the swap rate as a PV01-weighted combination of Libor forwards.

The CDS-LMM under default–rate independence (§5.1 of the paper):

$$dR_k(t) = \bar{\sigma}_k(t) R_k(t) d\bar{V}_k^k(t), \quad dF_k(t) = \sigma_k(t) F_k(t) dZ_k^k(t), \quad dV_i dZ_j = 0, \quad (7)$$

each rate a lognormal martingale under its own measure $\bar{Q}^k$ (for CDS) or Q^k (for Libor), with vanishing cross-correlation. The dynamics of R_i under a foreign measure $\bar{Q}^j$ (with $i \neq j$):

$$dR_i(t) = \bar{\sigma}_i(t) R_i(t) \sum_{h=(i \wedge j)+1}^{i \vee j} h_i^j \rho_{i,h} \frac{\bar{\sigma}_h(t) R_h(t)}{R_h(t) + \alpha_h/\text{LGD}} dt + \bar{\sigma}_i(t) R_i(t) dV_i^j(t), \quad (8)$$

with $h_i^j = \mathbf{1}_{j < i} - \mathbf{1}_{j > i}$. *Says:* the LMM machinery transplants intact to CDS spreads, modulo a factor $R_h + \alpha_h/\text{LGD}$ in the drift that comes from the defaultable bond ratio $\bar{P}(t, T_{h-1})/\bar{P}(t, T_h)$.

CMCDS pricing under drift-freezing (eq. at p. 21):

$$\text{CMCDS}_{a,b,c}(0) = \sum_{j=a+1}^b \alpha_j \bar{P}(0, T_j) \left[\sum_{i=j}^{j+c} \bar{w}_{ij}(0) R_i(0) e^{T_{j-1} \sigma_i \sum_{h=j+1}^i \rho_{i,h} \sigma_h R_h(0) / (R_h(0) + \alpha_h / \text{LGD})} - R_j(0) \right], \quad (9)$$

with $\bar{w}_{ij}(t) = \alpha_i \bar{P}(t, T_i) / \sum_{h=j}^{j+c} \alpha_h \bar{P}(t, T_h)$. Says: closed-form CMCDS valuation under (i) the drift-freezing approximation $\mu_i^j(R(t), t) \approx \mu_i^j(R(0), t)$, (ii) representation of the multi-period swap rate as a linear combination of one-period rates with PV01 weights, and (iii) deterministic vols and constant correlations.

3 Derivations & commentary

Commentary

The headline trick of the paper is the subfiltration $\mathcal{H}_t$ of default-free information, importing Jeanblanc-Rutkowski (2000) into a Jamshidian (2004)-style market-model construction. The problem in Schonbucher (1999) was that the defaultable-bond numeraire $\mathbf{1}_{\tau > t} \bar{B}_k(t)$ is not strictly positive — it collapses to zero on the default event — so the change-of-measure machinery only gives a measure absolutely continuous with Q , not equivalent. The paper's $C_{a,b}$ numeraire is the pre-default defaultable PV01, which is strictly positive everywhere; under $\mathcal{H}_t$ -conditioning $1/C_{a,b}(t)$ is bounded; the resulting $\bar{Q}^{a,b}$ is fully equivalent to Q .

Commentary

The conditional-independence assumption (Remark 4, paper p. 9) is non-trivial: it says that *the price of an $\mathcal{H}$ -measurable payoff depends only on $\mathcal{H}_t$, not on whether default has already occurred*. Jamshidian (2004) notes this “degrades the role of the subfiltration”. The financial reading: pre-default valuations of default-free claims don't depend on the fact of survival — sensible. With this, the Brigo (2005) Cox-process formula generalises to the full subfiltration framework.

Commentary

Why the term-structure model is hard. Eq. (5) expresses R_j as a ratio of forward survival probabilities under Q^j . So the defaultable bond *ratio* $\bar{P}(t, T_{j-1})/\bar{P}(t, T_j)$ involves $Q^{j-1}(\tau > T_{j-1} | \mathcal{H}_t) / Q^j(\tau > T_j | \mathcal{H}_t)$, a *cross-measure* probability ratio — not the same as $Q^j(\tau > T_{j-1} | \mathcal{H}_t) / Q^j(\tau > T_j | \mathcal{H}_t)$ that appears in the definition of R_j . Hence $\bar{P}(t, T_{j-1})/\bar{P}(t, T_j)$ cannot be expressed purely in terms of $R_j(t)$'s. This is the “measure mismatch” problem (paper p. 16).

Commentary

The mismatch disappears under independence of default and rates: then the survival probabilities are the same under all forward measures, and $\bar{P}(t, T_{j-1})/\bar{P}(t, T_j) = (1 + \alpha_j R_j(t) / \text{LGD})(1 + \alpha_j F_j(t))$ (paper p. 18). This finally allows a tractable CDS-LMM. The price is giving up the joint stochasticity of default-free interest rates and credit spreads — a major modelling concession, but acceptable in practice for vanilla CDS-derivative pricing.

Commentary

Empirical takeaways. Table on p. 11 shows DT, DCX, FT CDS options as of 26-Mar-2004: implied vols 50–55%, an order of magnitude higher than interest-rate swaptions but consistent with Schonbucher (2004) historical estimates. Sensitivity to recovery rate is small ($\pm 1\%$); sensitivity to a ± 50 bp zero-curve shift is also small (± 0.5 – 0.7%). *The implied vols are the right way to compare CDS options across strikes and maturities, not raw prices.*

Commentary

The CMCDS pricing (Sec. 5.2) is the killer application: a CMCDS pays the running $(c + 1)$ -period CDS spread at each coupon date, which combines a credit underlying with a maturity-rolling feature. Without a CDS term-structure model, this is hard; with the LMM-style construction here, it falls out via drift-freezing in closed form (9). The empirical example on Fiat shows the “convexity adjustment” (the difference between the naive valuation ignoring CDS-rate dynamics and the full model) is meaningful at typical vols and correlations.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- Pre-default defaultable bond $\bar{P}(t, T)$ and the par CDS spread (2).
- Defaultable PV01 $C_{a,b}(t)$ (3) and the corresponding measure $\bar{Q}^{a,b}$.
- Standard CDS option Black formula (4).
- One-period CDS spread $R_j(t)$ as a forward survival-probability ratio (5).
- Multi-period spread as weighted combination (6).
- CDS-LMM dynamics (7) and the drift under a foreign measure (8).
- Closed-form CMCDS price (9) under drift-freezing.

Canonical numerical cases:

- European CDS options on 3 corporates (Deutsche Telecom C_1 , Daimler Chrysler C_2 , France Telecom C_3) on 26-March-2004, recovery $\text{Rec} = 0.4$, expiry $T_a = 20\text{-Jun-2004}$ (and $T'_a = 20\text{-Dec-2004}$ for C_1), maturity $T_b = 20\text{-Jun-2009}$.
- Fiat CMCDS with $a = 0$, $b = 20$, $c = 20$ (resetting quarterly) on 20-Dec-2004, recovery $\text{Rec} = 0.4$.

Expected numerical outputs.

CDS options	(paper p. 11, mid prices and implied vols):
	$R_{a,b}(0)$ (bp) K (bp) Mid quote (bp) Implied $\bar{\sigma}_{a,b}$
Option C_1 on T_a	61 60 32.5 62.16%
Option C_2 on T_a	97.3 94 39 54.68%
Option C_3 on T_a	62.7 61 25 52.01%
Option C_1 on T'_a	65.4 61 35 51.45%

Recovery-rate sensitivity (C_1 on T_a , table p. 12): implied vol 50.02% at $\text{Rec} = 20\%$, 50.31% at $\text{Rec} = 40\%$, 50.90% at $\text{Rec} = 60\%$. *Very weak dependence.*

Curve-shift sensitivity (± 50 bp shift): C_1 on T_a : vol 49.68% at -50 bp, 50.31% at 0, 50.93% at $+50$ bp.

CMCDS Fiat convexity adjustment (table p. 22, $a = 0, b = 20, c = 20$):

	$\rho = 0.7$	$\rho = 0.8$	$\rho = 0.9$	$\rho = 0.99$
$\sigma = 0.1$	0.000659	0.000754	0.000848	0.000933
$\sigma = 0.2$	0.002662	0.003047	0.003435	0.003784
$\sigma = 0.4$	0.011066	0.012742	0.014442	0.015995
$\sigma = 0.6$	0.026619	0.030964	0.035464	0.039652

CMCDS participation rate $\phi_{0,20,20}(\sigma, \rho)$ (decreasing in both vol and correlation): from ~ 0.714 at low vol to ~ 0.599 at $\sigma = 0.6, \rho = 0.99$.

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(1)	$\text{CDS}\Pi_t$	CDS discounted payoff (tractable convention)
(2)	$R_{a,b}(t)$	par CDS spread
(3)	$C_{a,b}(t)$	defaultable PV01 (numeraire)
(4)	CDSOption_t	Black-formula CDS option price
(5)	$R_j(t)$	one-period CDS as forward survival-prob ratio
(6)	$R_{a,b}$ in terms of R_j	PV01-weighted aggregation
(7)–(8)	LMM-style dynamics	CDS spreads as lognormal martingales under natural measures
(9)	$\text{CMCDS}_{a,b,c}(0)$	closed-form CMCDS under drift-freezing

Invariants and identities to verify:

1. *Defaultable bond limit*: as $\text{Rec} \rightarrow 1$ (no loss), $R_{a,b}(t) \rightarrow 0$.
2. *Pre-default identity*: $\mathbf{1}_{\tau > t} \bar{P}(t, T) = \mathbb{E}^Q[D(t, T) \mathbf{1}_{\tau > T} | \mathcal{F}_t]$, so $\bar{P}(t, T) \leq P(t, T)$ always.
3. *$C_{a,b}$ positivity*: under $Q(\tau > t | \mathcal{H}_t) > 0$ a.s. (paper's assumption), $C_{a,b}(t) > 0$ pre-default (Proposition 7 of paper).
4. *Telescoping of one-period spreads to multi-period*: (6) holds identically, by construction of $C_{a,b}$ from $\bar{P}(t, T_i)$.
5. *ATM CDS option*: at $K = R_{a,b}(t)$, $d_1 = -d_2 = \bar{\sigma}_{a,b} \sqrt{T_a - t}/2$, and the option price is approximately $R_{a,b}(t) C_{a,b}(t) \bar{\sigma}_{a,b} \sqrt{T_a - t} / \sqrt{2\pi}$.
6. *Implied vol back-out*: given a market quote (e.g. C_1 : mid 32.5 bp on $R_{a,b}(0) = 61$, $K = 60$, $T_a - t \approx 0.24Y$), solving the Black formula (4) for $\bar{\sigma}_{a,b}$ returns 62.16%.
7. *LMM consistency at $\rho = 0$* : in (8), all $\rho_{i,h} = 0$ collapses the drift to zero, recovering the martingale dynamics under $\bar{Q}^j = \bar{Q}^i$.
8. *CMCDS at $\sigma = 0$* : setting all $\bar{\sigma}_i \rightarrow 0$ in (9) collapses the exponential to 1 and gives the no-convexity-adjustment value $\sum \alpha_j \bar{P}(0, T_j) [\sum_i \bar{w}_{ij} R_i(0) - R_j(0)]$. Reproduces the $\rho = 0$ rows of the participation-rate table (since with $\sigma = 0$ the only correction is from the LMM-style replacement of multi-period rates).
9. *Conditional-independence test*: under independence of rates and default, the one-period spread becomes $R_j(t) = (\text{LGD}/\alpha_j) [Q(\tau > T_{j-1} | \mathcal{H}_t) / Q(\tau > T_j | \mathcal{H}_t) - 1]$ with Q in place of $\bar{Q}^j$. Verify this equals (5) when F_j and survival are uncorrelated.

5 Glossary of symbols (paper's notation)

τ	Default time
T_a, T_b	Start and maturity of CDS premium leg
α_i	Year fraction of period $(T_{i-1}, T_i]$
LGD	Loss given default (= 1 – recovery)
R	Running CDS premium (rate) paid by protection buyer
$R_{a,b}(t)$	Par CDS spread for maturity $T_a \rightarrow T_b$
$D(t, T)$	Stochastic bank-account discount B_t/B_T
$P(t, T)$	Default-free zero-coupon bond price
$\bar{P}(t, T)$	Pre-default defaultable bond price
Q	Risk-neutral measure (bank-account numeraire)
Q^j	T_j -forward measure under $P(t, T_j)$
$\bar{Q}^{a,b}$	Measure under defaultable-PV01 numeraire $C_{a,b}$
$Q(\tau > t \mathcal{H}_t)$	Conditional survival probability (strictly positive)
$\mathcal{F}_t$	Full filtration (default time observable)
$\mathcal{F}_t^\tau$	Subfiltration generated by τ
$\mathcal{H}_t$	Default-free subfiltration
$C_{a,b}(t)$	Defaultable PV01 numeraire
$R_j(t) = R_{j-1,j}(t)$	One-period forward CDS spread
$F_k(t)$	Default-free forward Libor (LMM variable)
$\bar{\sigma}_{a,b}$	Implied volatility of $R_{a,b}$ under $\bar{Q}^{a,b}$
$\rho_{i,h}$	Correlation between Brownians of R_i, R_h
h_i^j	Sign indicator: $\mathbf{1}_{j < i} - \mathbf{1}_{j > i}$
$\bar{w}_{ij}(t)$	PV01 weight $\alpha_i \bar{P}(t, T_i) / \sum_h \alpha_h \bar{P}(t, T_h)$
Black(F, K, v)	Standard Black formula

TODO / Open questions

- Reproduce the implied-vol back-out on the four worked CDS options (Deutsche Telecom, Daimler Chrysler, France Telecom, with the T_a, T'_a variants). Target: 62.16%, 54.68%, 52.01%, 51.45% (paper p. 11).
- Verify the weak dependence of $\bar{\sigma}_{a,b}$ on recovery rate (variation $\leq 0.5\%$ over $\text{Rec} \in [0.2, 0.6]$) and the comparable weak dependence on parallel curve shifts.
- Implement the CDS-LMM under default-rate independence (7) with vanishing cross-correlation; propagate one-period rates $R_k(t)$ and Libor rates $F_k(t)$ in their own natural measures.
- Reproduce the Fiat CMCDS convexity-adjustment table ($a = 0, b = 20, c = 20, \text{Rec} = 0.4$) across the (σ, ρ) grid; target the exact values in (9).
- Reproduce the participation-rate table (decreasing in σ, ρ).
- Implement the change-of-measure drift adjustment (8) and Monte-Carlo it against the drift-freezing closed form for a CMCDS; quantify the approximation error.
- Cross-check against Zhou (2008) (zhou_2008_cds): Brigo-Morini give the spread as a forward-survival-probability ratio (eq. 13), Zhou gives the bond-implied CDS as a weighted sum of bond coupon, Libor and bond discount; at consistent inputs they should agree on the par-spread level.